

END TERM EXAMINATION

SIXTH SEMESTER [B.TECH] MAY-JUNE 2025

Paper Code: CIE-374T
ECE-318T
AI-302T

Subject: Artificial Intelligence

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit. Assume suitable missing data if any.

Q1 Attempt any five

(5x5=25)

- (a) What do you mean by AI & briefly explain its types?
- (b) Define various kind of agents and their function.
- (c) Briefly explain the knowledge representation methods.
- (d) Explain MINMAX Algorithm.
- (e) What is Machine Learning?
- (f) What is PEAS representation?
- (g) Define Problem as a State Space Search.

UNIT-I

- Q2 (a) Explain DFS & BFS in detail. (5.5)
(b) Explain A* algorithm with suitable example. (7)

- Q3 (a) Trace the constraint satisfaction procedure for solving the following cryptarithmic problem. (7.5)

D O N A L D
+ G E R A L D

R O B E R T

- (b) Differentiate Uniformed Search and Informed Search. (5)

UNIT-II

- Q4 (a) Explain Resolution Principle with example. (6.5)
(b) Write short note with example on:
i. Symbolic Reasoning.
ii. Non-monotonic Reasoning.
iii. Statistical Reasoning (6)

- Q5 (a) Differentiate between Inductive and Deductive Reasoning. (5)
(b) What is the difference between forward chaining and backward chaining? (5)
(c) Define Proposition Logic. (2.5)

UNIT-III

- Q6 (a) Explain the concept of Alpha-Beta pruning with example. (7.5)
(b) Explain in detail about Explanation-based learning module. (5)

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- Q7 (a) Define NLP and its types. (7.5)
(b) Difference between Neural network learning and genetic learning. (5)

UNIT-IV

- Q8 (a) Describe the architecture of Expert Systems in detail. (7.5)
(b) Explain the concept of Bayesian network in representing knowledge in an uncertain domain. (5)
- Q9 (a) What do you understand by K-means clustering algorithm? (5)
(b) Difference between Supervised and Unsupervised learning. (5)
(c) Define Fuzzy Logic System. (2.5)

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